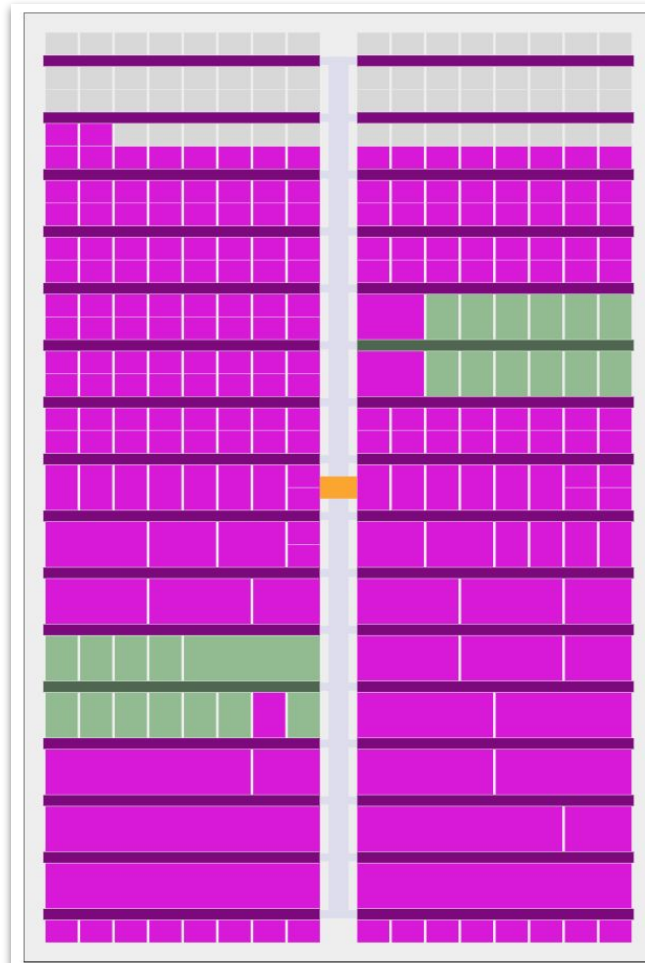


The background of the slide is a grayscale image of a complex circuit board, possibly a tapeout, with a grid of vertical and horizontal lines overlaid. The grid is composed of thin white lines that create a series of rectangular cells across the entire image. The circuit board itself is dark with intricate patterns of lines and components visible.

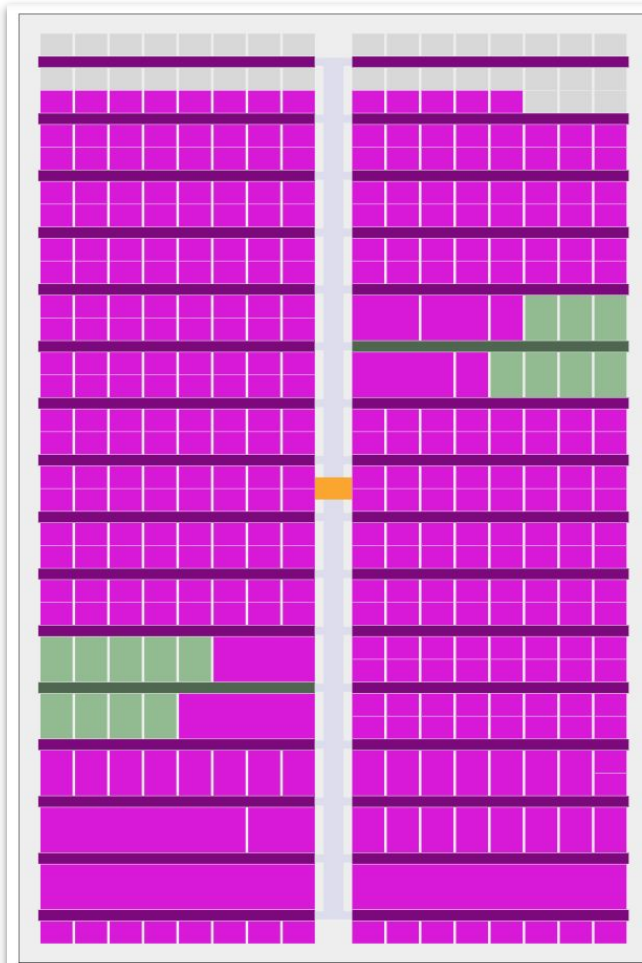
Tiny Tapeout

FSiC 2025 Matt Venn

- [10+ chips](#)
- 30+ countries
- 50+ universities
- 100+ [silicon proven](#)
- 1500+ designs
- 5000+ people



TT06



TT09



Danish Chips
Competence
Centre



synopsys

SyoSil
Systems on Silicon

<https://tinytapeout.com/credits/>



Danish Chips
Competence
Centre



synopsys

SyoSil
Systems on Silicon

<https://tinytapeout.com/credits/>



Danish Chips
Competence
Centre



synopsys

SyoSil
Systems on Silicon

<https://tinytapeout.com/credits/>

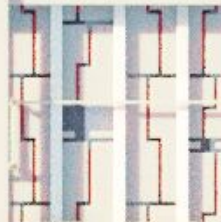
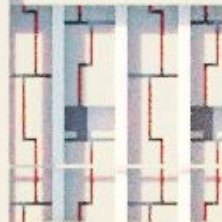
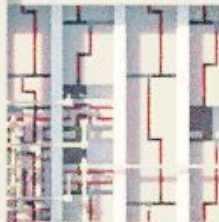
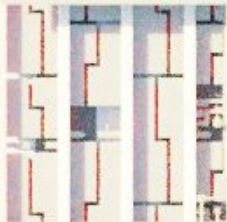
“

Tiny Tapeout is the most valuable project we have in the semiconductor industry for workforce development and innovation...

Jørgen Kragh Jakobsen - ASIC industry professional.

Tiny Tapeout is genius! It allows students, hobbyists, and professionals to easily create custom silicon circuit blocks, supporting digital, analog, or mixed-signal... indispensable for modern IC design education!

Prof. Harald Pretl - Senior Principal Engineer at Intel, JKU, Research Institute for Integrated Circuits.





htamas  25/06/2025, 19:04

Congrats to the 3000th member 🎉



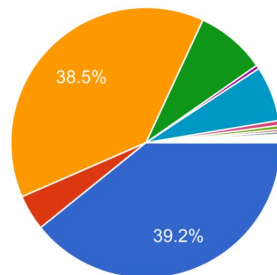
Tiny Tapeout  >

3,000 Members • Community



What's your region?

681 responses

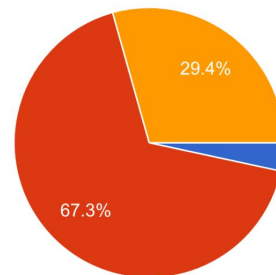


- North America
- Central / South America
- Europe
- India
- Africa
- Asia
- Australia
- Oceania

▲ 1/3 ▼

What's your age?

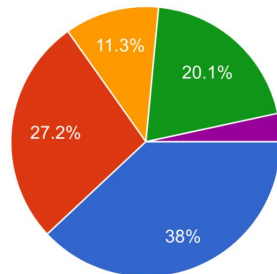
681 responses



- < 20
- 20 to 40
- > 40

What's your background?

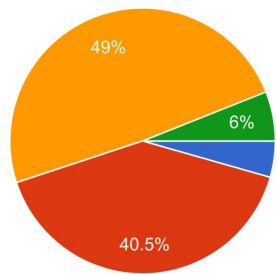
681 responses



- Working in industry
- Academic - student
- Academic - professor
- Hacker / maker
- Just curious

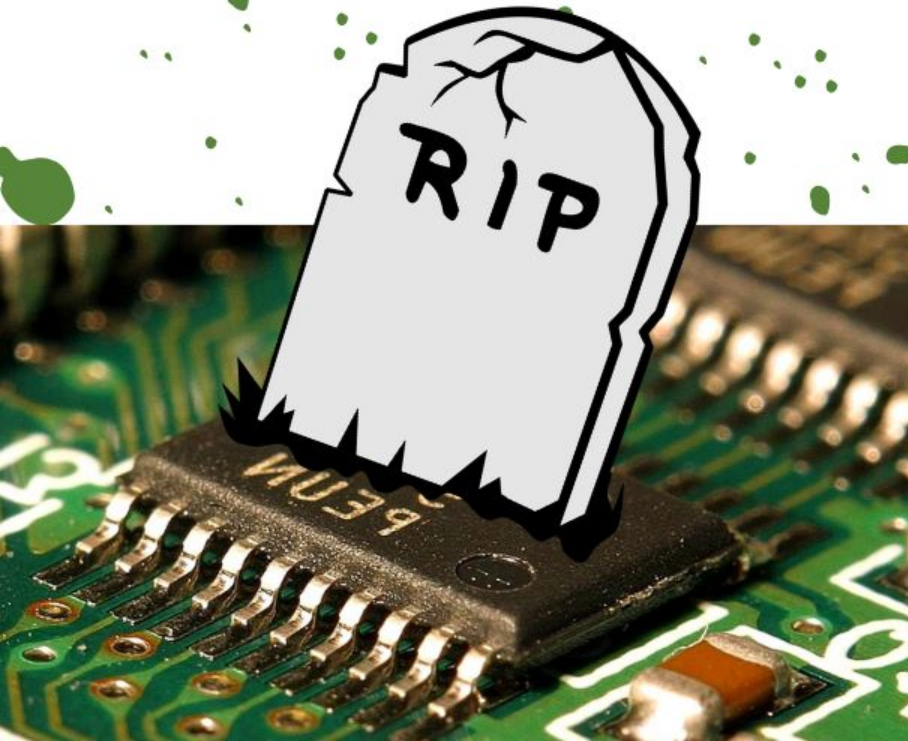
How many designs do you expect to submit to Tiny Tapeout in the next 2 years?

681 responses

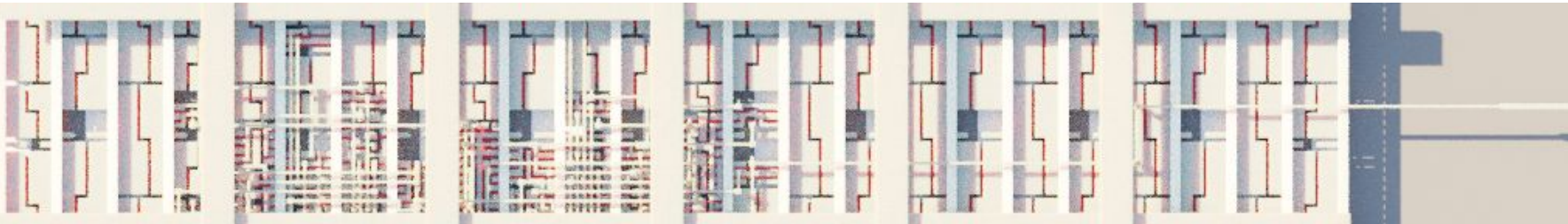


- 0
- 1
- 2 to 5
- 6 or more

EFABLESS



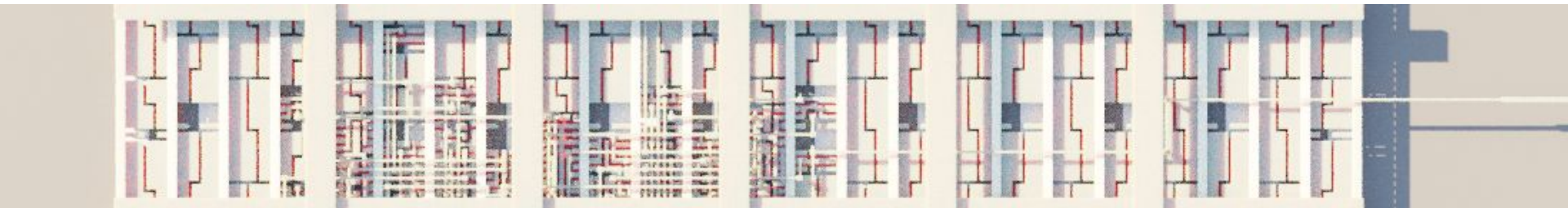
- People often ask about products, IMO that's still a long way out.
- We need to develop high quality IPs like SRAM.
- We need reliable, regular tapeouts.
 - Fund entities that create and maintain open source PDKs on mature nodes.
 - Make free open source chip fabrication possible for every student in Europe through public funding for several runs of chip production.
 - Sign the open letter: <https://open-source-chips.eu/>

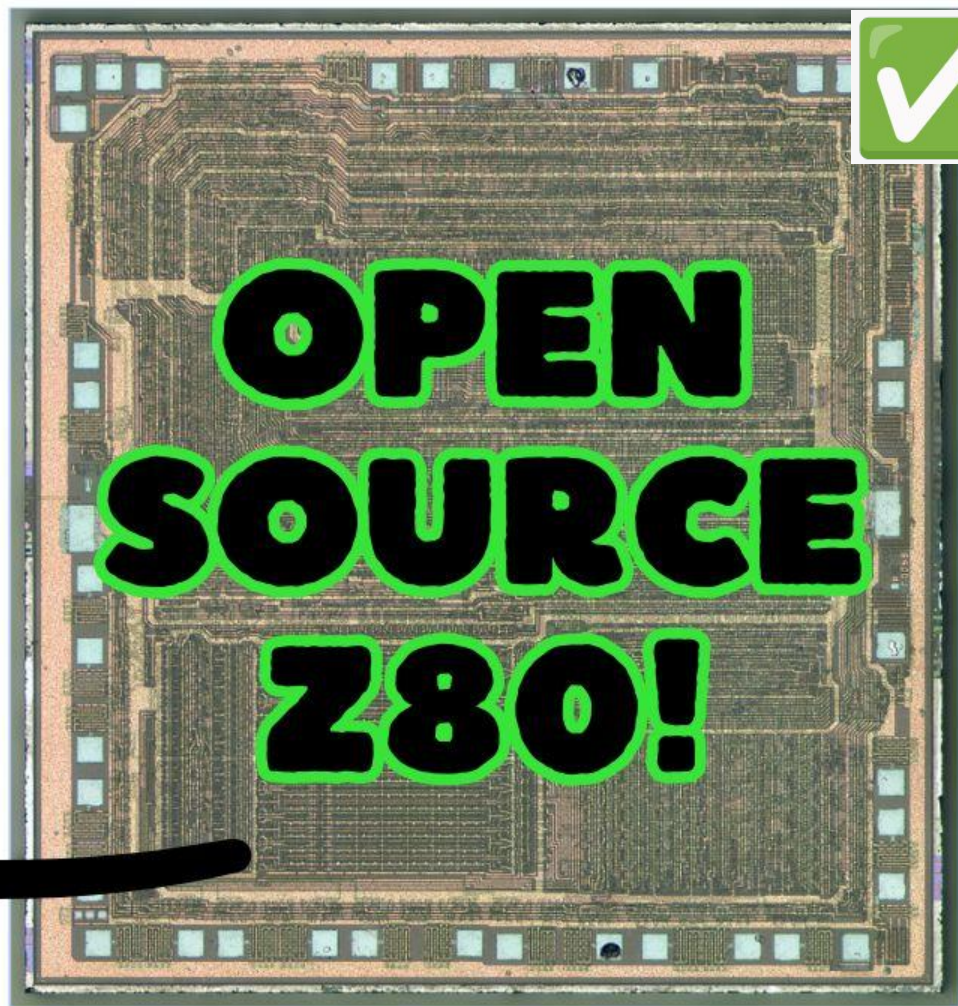
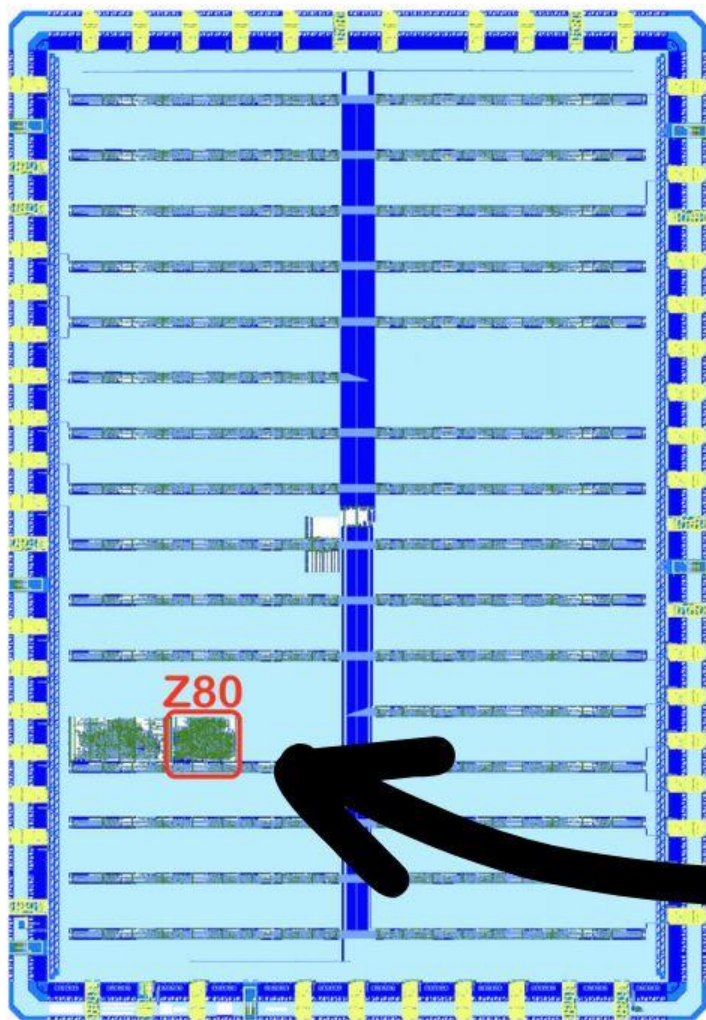


<https://tinytapeout.com/runs/silicon-proven/> ~20% tested

#	Shuttle	Title	✓
0	TT06	KianV uLinux SoC	5
1	TT05	Flappy VGA	5
2	TT05	VGA Dino Game	5
3	TT05	Simon Says game	4
4	TT05	VGA clock	4
5	TT05	Game of Life 8x32 (siLife)	4
6	TT07	DVD Screensaver with Tiny Tapeout Logo (Tiny VGA)	3
7	TT07	Zilog Z80	3
8	TT06	PILIPINAS	3

- Community is vital, not just for Tiny Tapeout, but open source in general.
- Taking it for granted leads to burn out, abuse.
- We need to support diversity of people and thought, while challenging damaging behaviour - it's not easy.
- Join the server: <https://fossi-chat.org/>

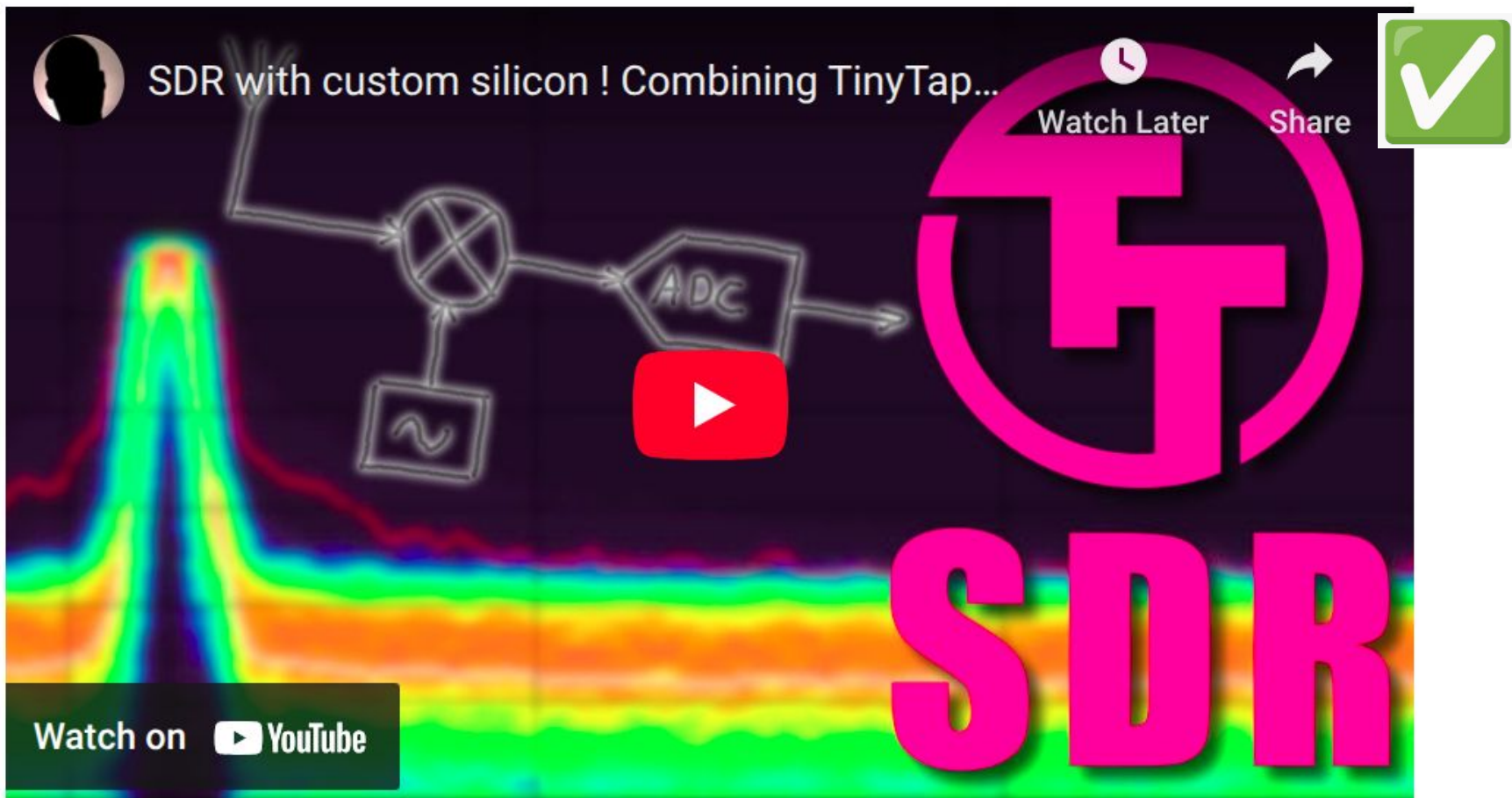




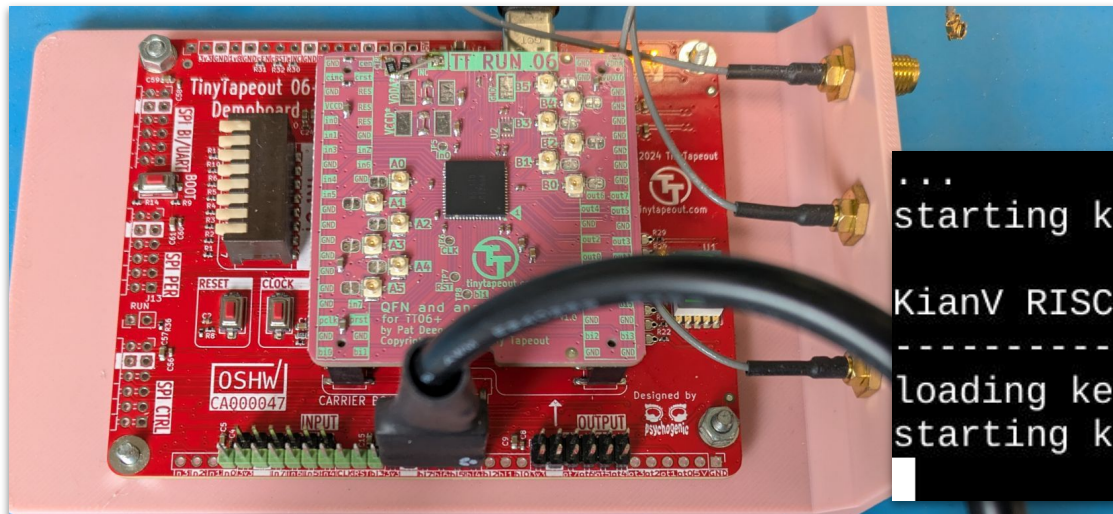
12 bit SAR ADC - 3D



12 bit ADC



If proof were needed that Tiny Tapeout delivers functional hardware, here it is: a basic SDR built using chips from the open silicon fabrication program. (👤: Sylvain Munaut)



```
...  
starting kernel
```

```
KianV RISC-V uLinux ASIC Tiny Tapeout SOC
```

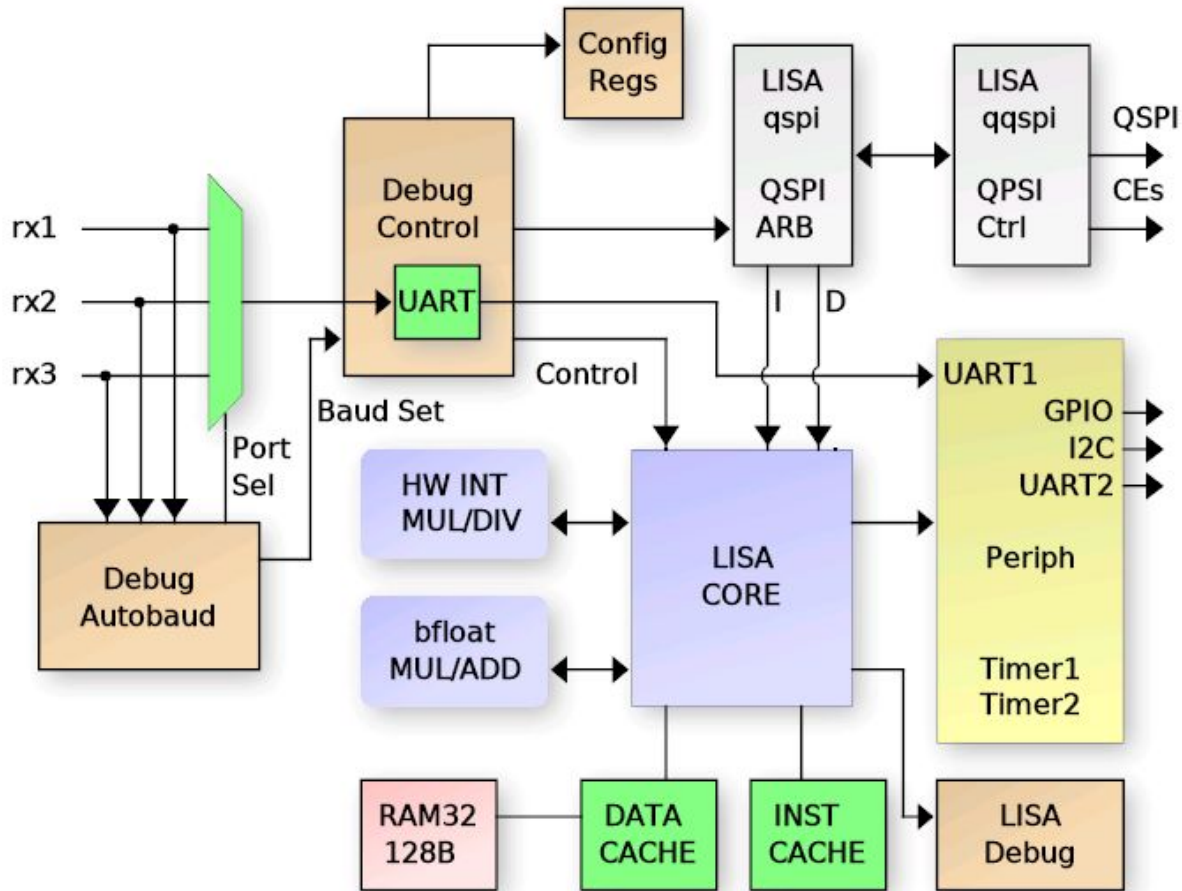
```
-----  
loading kernel Image from flash .....  
starting kernel
```

```
starting pid 32, tty '/dev/console':  
/sh console 0 vt100'
```

```
/ # cat /proc/uptime  
149.08 33.09
```

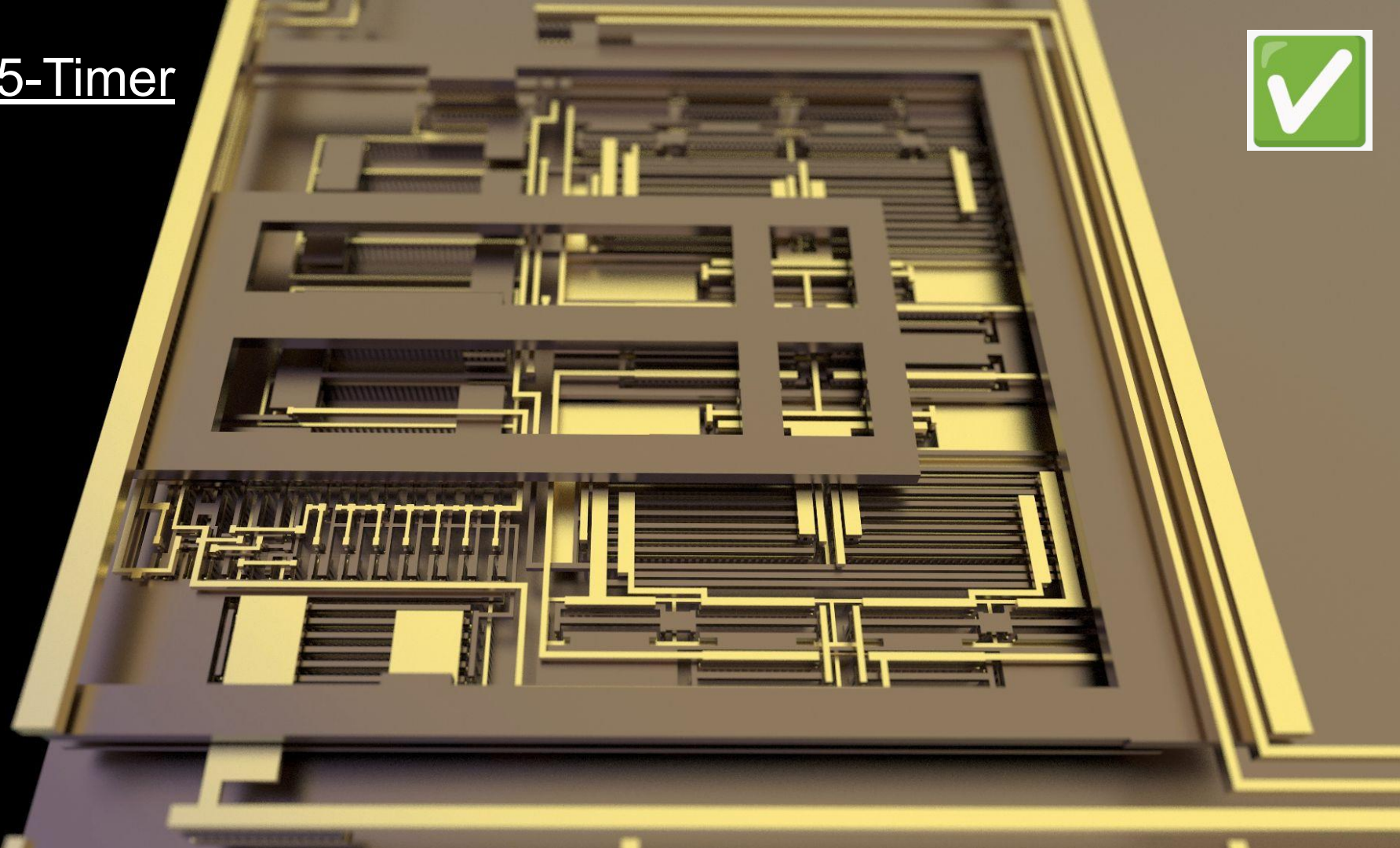


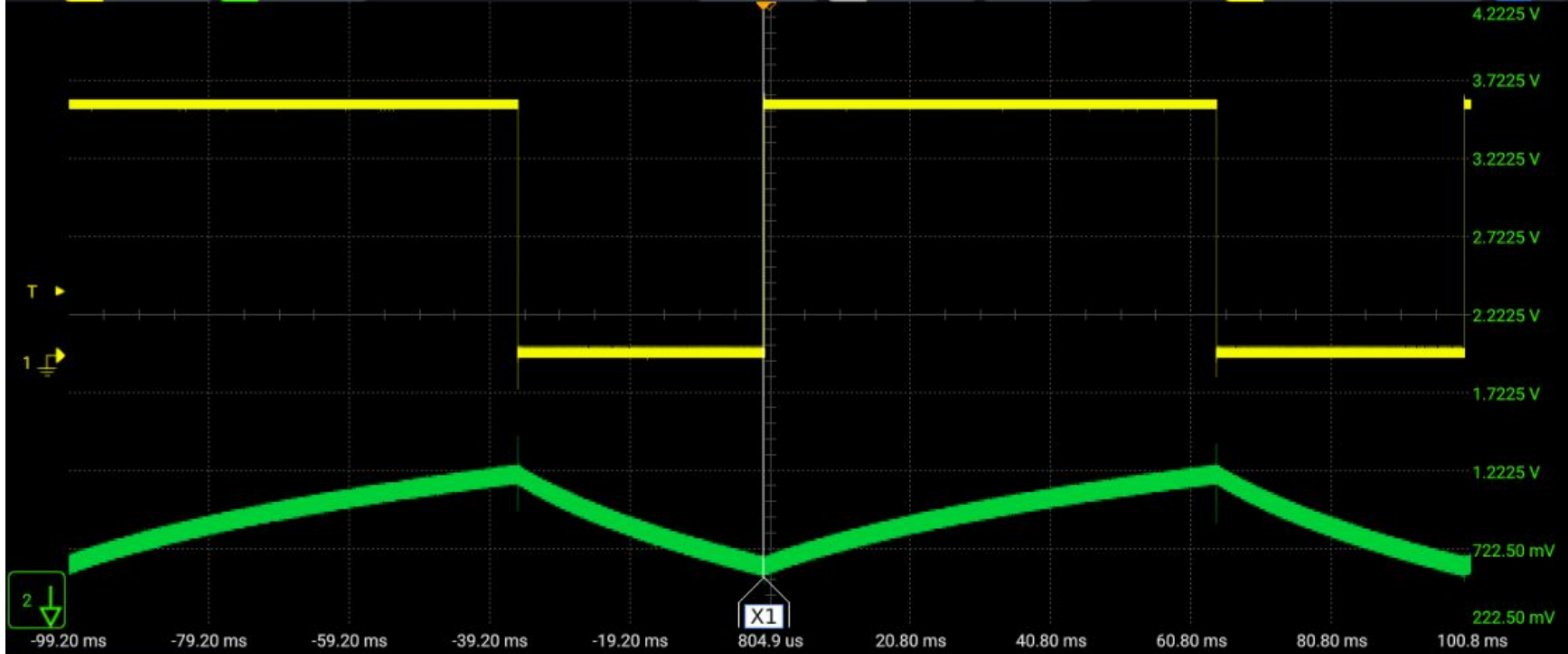
https://tinytapeout.com/runs/tt06/tt_um_kianV_rv32ima_uLinux_SoC



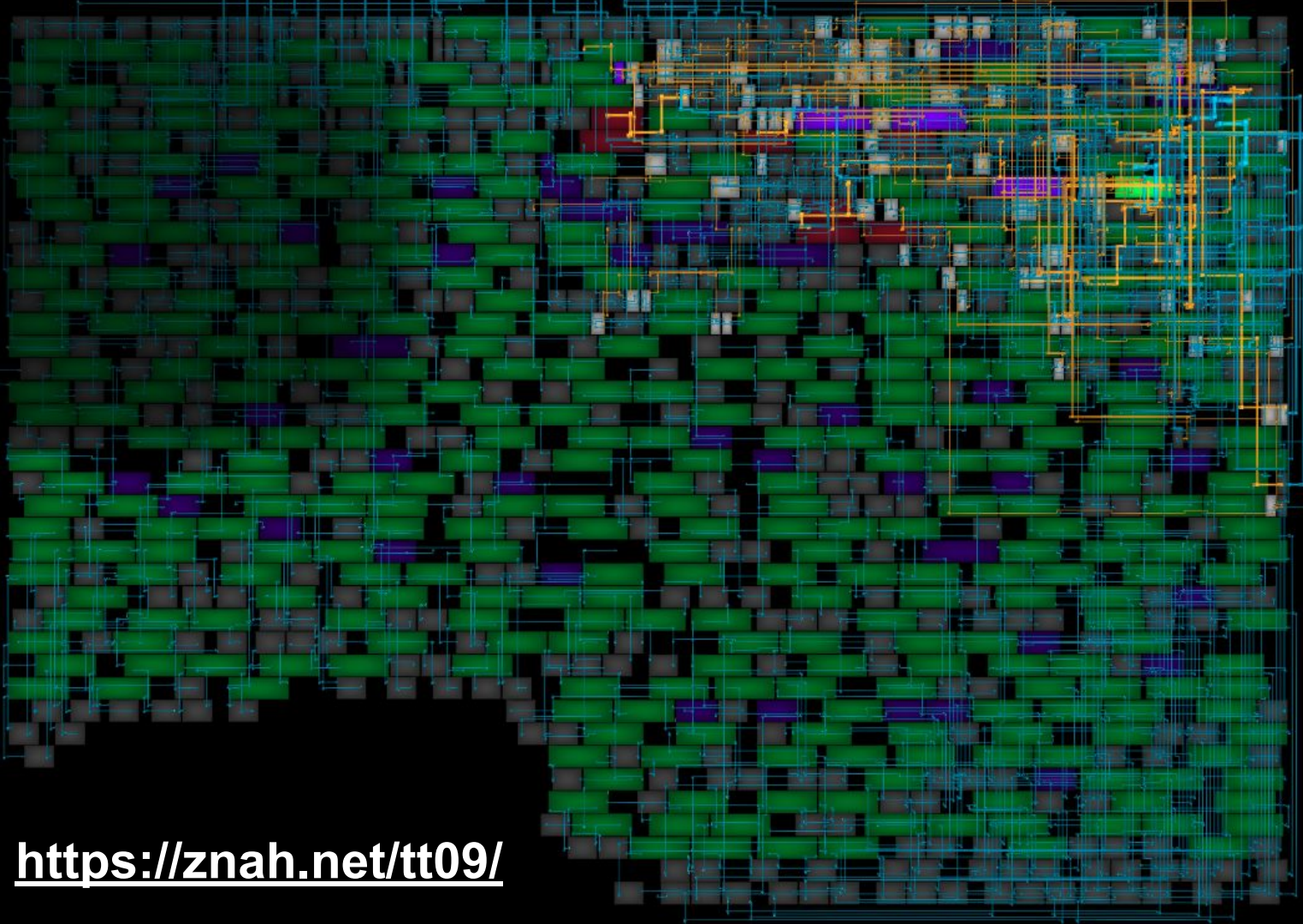
LISA microcontroller https://tinytapeout.com/runs/tt06/tt_um_lisa

555-Timer





Measurements		Markers					
Measurement	Current	Mean	Min	Max	Std Dev	Count	
Freq(1)	10.022 Hz	18.430 kHz	9.7432 Hz	3.1130 MHz	113.76 kHz	866	
Ampl(1)	3.1250 V	1.0686 V	101.6 mV	3.3125 V	1.3927 V	930	
Rise(1)	<4.7 us	89.355 us	100 ns	2.4616 ms	213.16 us	626	
Fall(1)	<4.7 us	100.07 us	100 ns	1.6345 ms	232.55 us	627	



<https://znah.net/tt09/>

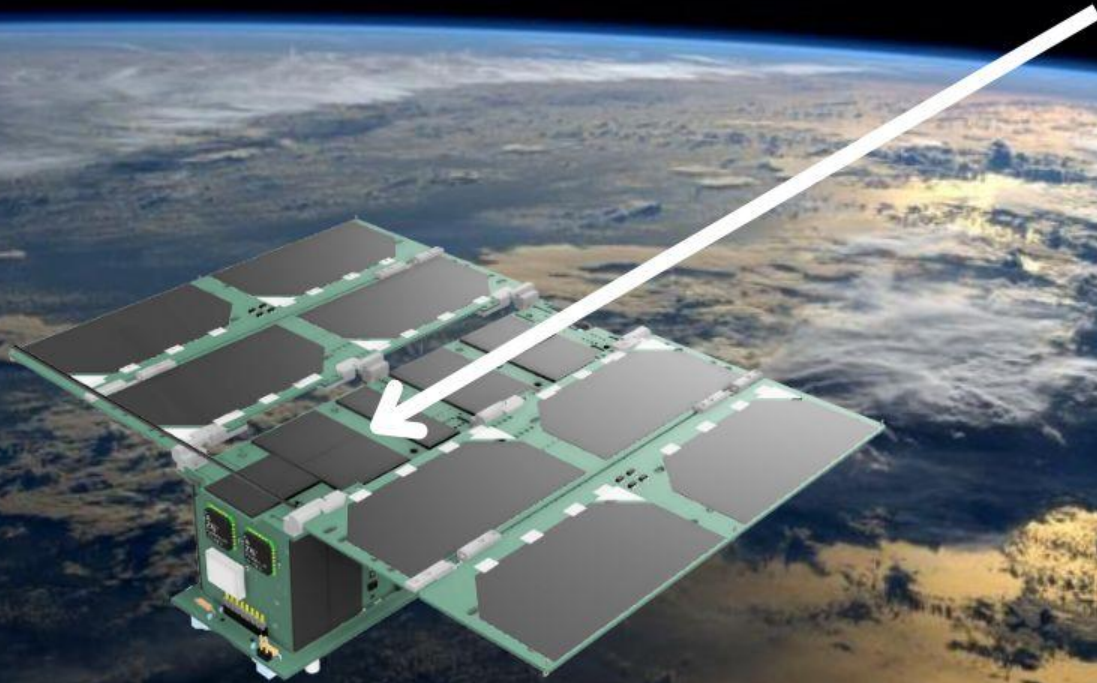
<http://tinytapeout.com/decap/tt07>

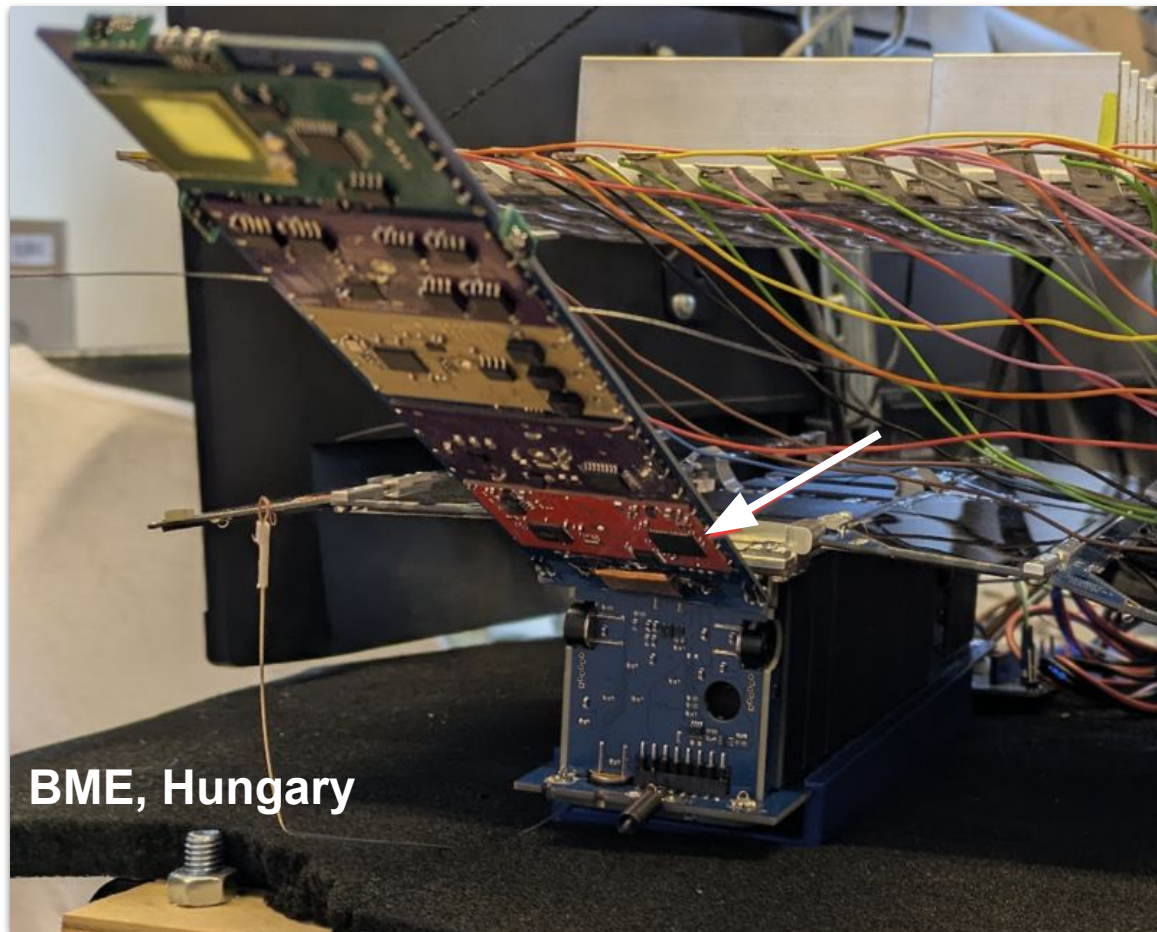


Texplained

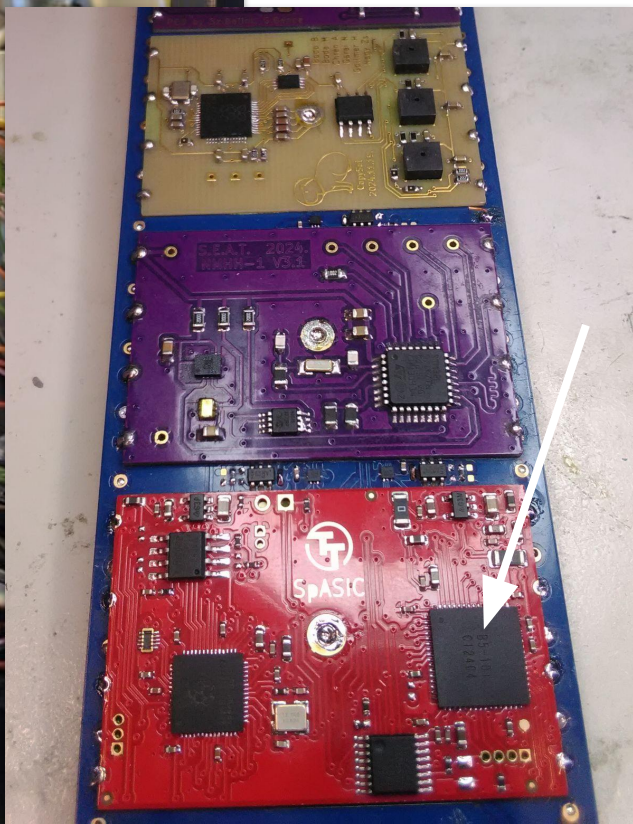
HARDWARE SECURITY INSIGHT

Your chip in Space!



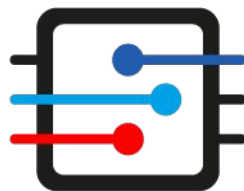


BME, Hungary





ETH zürich



SwissChips



IHP25b

Closes in

61

DAYS

23

HOURS

38

MINS

58

SECS

Tiles
49% left



PCBs
98% left



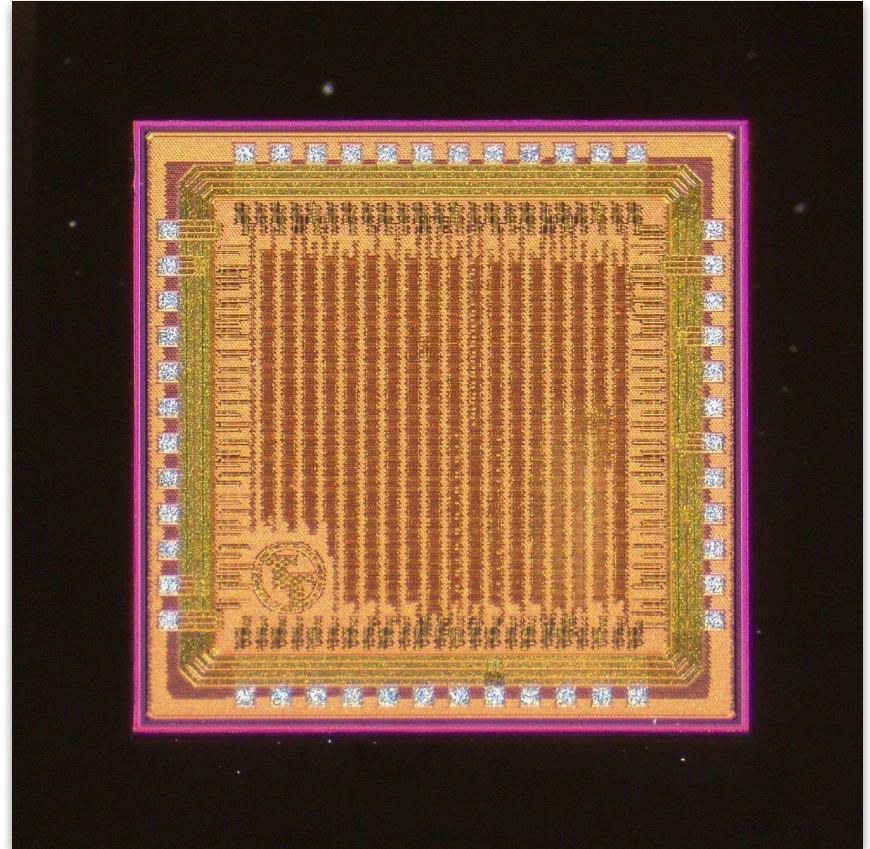
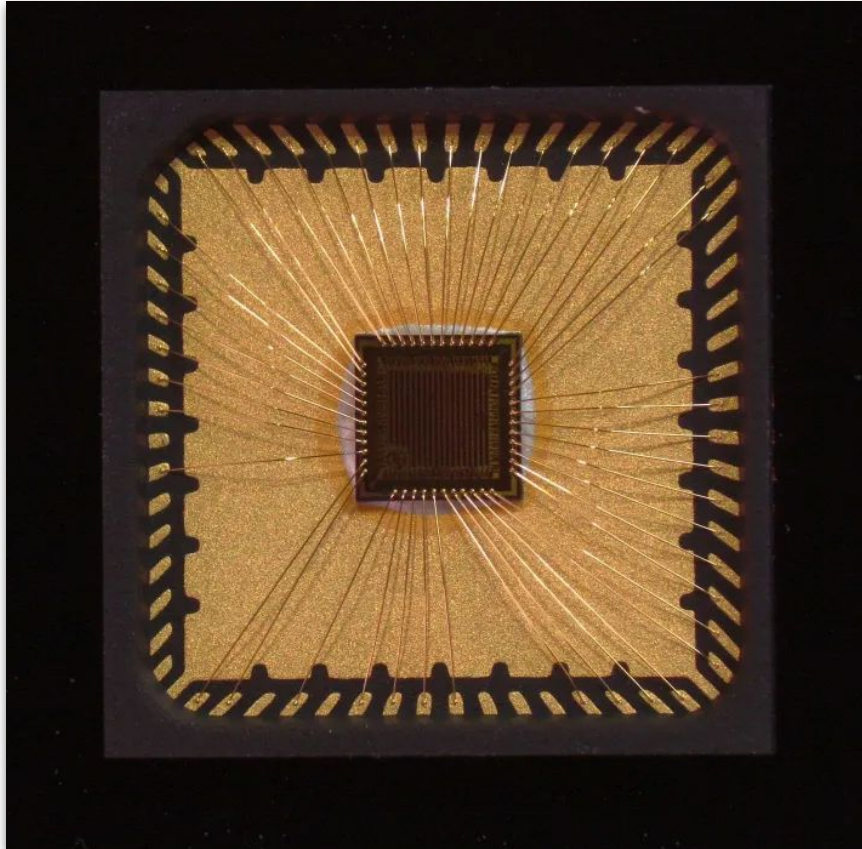
Dissemination and outreach engagement

- SwissChips Website: swisschips.ethz.ch
 - News, dissemination of research results (papers, presentations, demonstrators, etc.), chip gallery
 - SwissChips is also present on LinkedIn
- **Tiny Tapeout shuttles for SwissChips**
 - Chip design for everyone!
 - First workshop last Friday



Landesmuseum Zürich.

IHP 0p1 bringup success!





SKY25a
Closes in

75

DAYS

23

HOURS

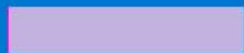
38

MINS

39

SECS

Tiles
99% left



PCBs
100% left





wafer.space

143 Cecil Street,
#17-04, GB Building,
Singapore 069542

info@wafer.space

18th June 2025

RE: Adapting Tiny Tapeout for GF180MCU letter of support

To Whom it May Concern,

This letter is to show wafer.space's support for adapting Tiny Tapeout's infrastructure to the open source GF180MCU technology manufactured by GlobalFoundries.



Millbrook, NY

Mathew Venn
Tiny Tapeout
Comunidad Valenciana, Valencia, Spain
Re: Letter of Intent

June 12, 2025

Dear Matthew:

This letter is to express the intent from the IEEE Electron Devices Society EDS to use Tiny Tapeout in the GF 180nm open PDK as a platform to train our members on the Design of Integrated Circuits. We strongly support the porting of the Tiny Tapeout infrastructure into



Synopsys, Inc.
675 Almanor Ave
Sunnyvale, CA 94085
T 650-584-5000

June 18, 2025

Mathew Venn
Tiny Tapeout
Comunidad Valenciana, Valencia, Spain

Subject: Letter of Intent

This letter is to express the intent from Synopsys Academic & Research Alliances to use Tiny Tapeout in the GF 180nm open PDK as a platform to train the next generation of students in chip design.

GF180 support is in progress!

1: Closing the submission loop with a minimum viable chip.

Submit to either wafer.space or Synopsys sponsored shuttle September 2025.

2: Minimum viable Tiny Tapeout chip with digital designs only.

Submit to wafer.space shuttle Oct-Nov 2025.

We are looking for funding to add analog and mixed signal support.

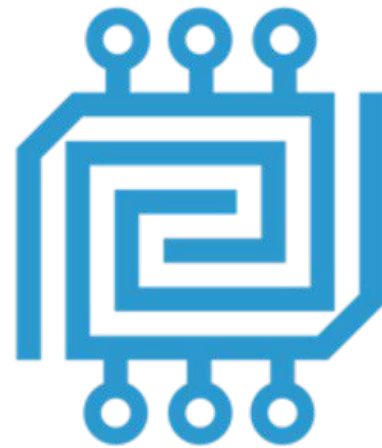


tillit



GlobalFoundries®

Join us in my home town for OrConf 2025!



September 12-14
Valencia Spain

Connect with me!

- Slides bit.ly/fsic-tt-2025 CC-BY 4.0
- [Newsletter](#)
- [Youtube channel](#)
- [Linked.in](#) / [Mastodon](#) / [Bluesky](#)
- Cite the Tiny Tapeout [IEEE paper](#)

